

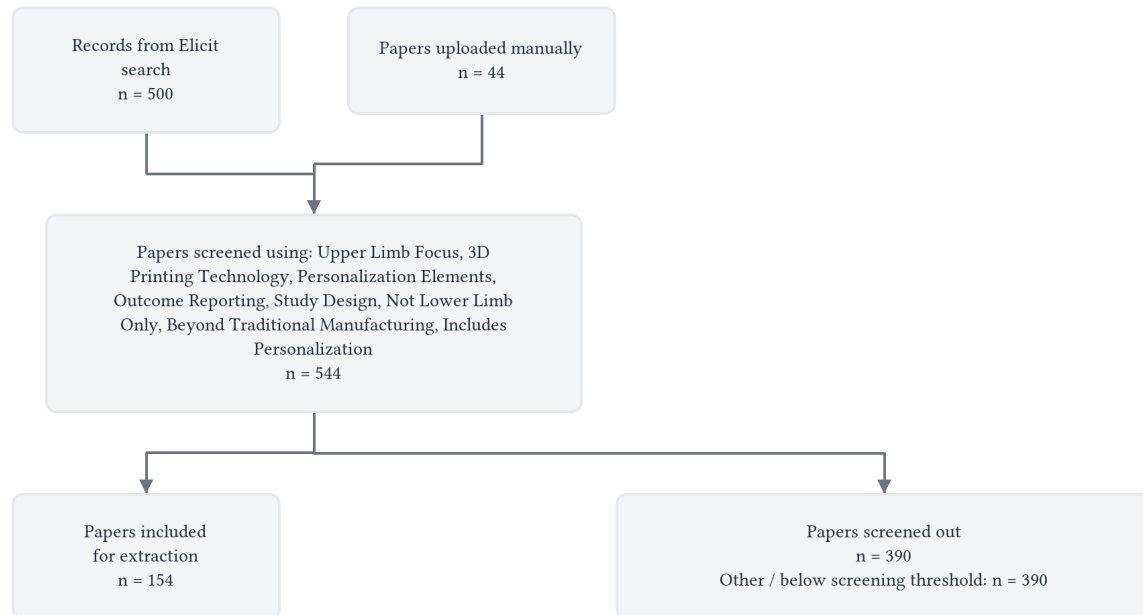
PARAMETRIZAÇÃO E IMPRESSÃO EM 3D DE PRÓTESES DE MEMBROS SUPERIORES UTILIZANDO DADOS ANTROPOMÉTRICOS E INTELIGÊNCIA ARTIFICIAL PARA CRIAR DISPOSITIVOS ALTAMENTE PERSONALIZADOS E FUNCIONAIS.

Parametric 3D printing driven by anthropometric data and AI-based EMG control can produce upper limb prostheses that are highly personalized in fit and functionally effective for gross prehension tasks at a fraction of the cost of commercial devices, but fine motor capability and long-term durability remain unresolved, and the approach is best validated for populations in low-resource settings who lack access to conventional prosthetics rather than as a direct replacement for advanced commercial myoelectric systems.

Abstract

Parametric design combined with additive manufacturing has produced upper limb prostheses and orthoses that are substantially more affordable than traditional alternatives, with material costs for most devices falling between USD 12 and USD 350 compared to USD 4,000–20,000 for commercial myoelectric prostheses [1–4]. Anthropometric personalization is achieved through methods ranging from manual tape-measure assessment and smartphone photogrammetry to 3D laser scanning integrated with parametric CAD environments, with the most rigorous implementations driving geometric parameters directly from scan data at sub-millimeter tolerances [5–7]. Genuine multi-dimensional parametrization—where independent geometric drivers are adjusted from patient measurements—demonstrably outperforms uniform scaling in matching natural digit kinematics and socket fit [7, 8], though most open-source community designs still rely on the simpler uniform-scaling approach [7]. AI-based EMG classifiers (SVM, KNN, artificial neural networks) achieve motion classification accuracies of 94–95% in controlled conditions [9, 10], and more advanced systems incorporating convolutional neural networks for grasp selection further automate device behavior [11]; however, high classification accuracy does not guarantee fine motor task completion, as evidenced by a 22.2% fine motor success rate alongside 100% gross motor success in one nine-participant evaluation [12]. Functional gains are consistently reported in gross prehension, grip strength for everyday objects, and activities of daily living, with standardized assessments such as the BAM-ULA, AM-ULA, Bilan 400, and MRC scale documenting meaningful improvements [12–15], yet grip forces remain below those of conventional body-powered prostheses [3] and fine manipulation remains a persistent gap. The evidence is clearest for populations in low-resource settings and for children with congenital or traumatic limb differences who would otherwise have no prosthesis; for these users, 3D-printed personalized devices yield substantial functional and psychosocial benefit [13, 16]. For adults replacing commercial myoelectric devices, the functional equivalence claim is not yet supported by the available clinical validation, which remains largely limited to case reports and small uncontrolled series, and is further constrained by unresolved durability concerns [4, 17, 18] and the absence of standardized manufacturing and regulatory protocols [16, 19].

Flow Diagram



Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "PARAMETRIZAÇÃO E IMPRESSÃO EM 3D DE PRÓTESES DE MEMBROS SUPERIORES UTILIZANDO DADOS ANTROPOMÉTRICOS E INTELIGÊNCIA ARTIFICIAL PARA CRIAR DISPOSITIVOS ALTAMENTE PERSONALIZADOS E FUNCIONAIS."

The search returned 500 total results from Elicit.

We retrieved 500 papers most relevant to the query for screening.

44 sources were uploaded to Elicit for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Upper Limb Focus:** Does this study focus on upper limb prostheses (arm, forearm, hand, or finger prostheses)?
- **3D Printing Technology:** Does this study involve 3D printing/additive manufacturing technologies for prosthetic production?

- **Personalization Elements:** Does this study incorporate anthropometric data collection and application in prosthetic design OR utilize parametrization, computational design, or artificial intelligence/machine learning in prosthetic development?
- **Outcome Reporting:** Does this study report on functional outcomes, fit assessment, personalization metrics, or usability of prosthetic devices?
- **Study Design:** Is this study an experimental study, observational study, case study, technical report, systematic review, or meta-analysis?
- **Not Lower Limb Only:** Is this study NOT focusing exclusively on lower limb prostheses?
- **Beyond Traditional Manufacturing:** Is this study NOT limited to traditional prosthetic manufacturing without 3D printing components?
- **Includes Personalization:** Does this study include anthropometric data or personalization elements (i.e., is it NOT a study on prosthetic devices that completely lacks anthropometric data or personalization elements)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

At abstract screening, the number of papers excluded for each primary reason was:

- **Other / below screening threshold:** n = 390

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **3D Printing Technology:**

Extract all details about 3D printing methods and technology including:

- Type of 3D printing technology used (FDM, SLA, SLS, etc.)
- Materials used for printing (PLA, ABS, TPU, etc.)
- Printer specifications and capabilities
- Printing parameters (layer height, infill density, print speed)
- Post-processing methods required
- Manufacturing time and complexity

- **Anthropometric Data Application:**

Extract information about how anthropometric data is collected and used including:

- Methods of anthropometric data collection (manual measurement, 3D scanning, photogrammetry, etc.)
- Specific body measurements taken (dimensions, circumferences, lengths)
- How measurements are converted to prosthetic design parameters
- Accuracy and precision of measurement methods
- Integration with design software or CAD systems

- **AI and Personalization Methods:**

Extract all artificial intelligence and personalization approaches including:

- Type of AI/ML algorithms used (neural networks, machine learning models, etc.)
- How AI is used in the design process (automated parametrization, optimization, etc.)
- Personalization features implemented (custom fit, functional adaptation, aesthetic preferences)

- Training data used for AI models
- Validation methods for AI-generated designs

- **Functional Performance:**

Extract functional outcomes and performance measures including:

- Types of prosthetic functions tested (grip strength, range of motion, dexterity tasks)
- Quantitative performance metrics and results
- Comparison with traditional prosthetics or baseline measurements
- User functional assessments and standardized tests used
- Specific tasks or activities evaluated
- Duration of functional testing

- **User Experience:**

Extract user acceptance, satisfaction, and experience data including:

- User satisfaction scores or ratings
- Comfort and fit assessments
- Aesthetic acceptance and preferences
- Learning curve and adaptation time
- Daily usage patterns and duration
- User feedback on personalization features
- Abandonment rates or continued use

- **Cost and Accessibility:**

Extract economic and accessibility information including:

- Manufacturing costs compared to traditional prosthetics
- Time to manufacture and deliver
- Required equipment and infrastructure
- Accessibility to different populations (geographic, economic)
- Scalability of the approach
- Training requirements for practitioners

- **Technical Limitations:**

Extract challenges, limitations, and technical issues including:

- Material limitations and durability issues
- Design constraints of 3D printing technology
- Accuracy limitations in fit or function
- Technical failures or problems encountered
- Maintenance requirements
- Integration challenges with existing healthcare systems
- Regulatory or approval challenges

Report

Due to the limitations of the AI model, we are only able to process 80 sources while writing a report. This report was written using the 80 sources that had the highest screening scores out of the 154 sources that we screened in and extracted data from.

Results

Characteristics of Included Studies

The 80 sources included in this review span primary development studies, case reports, literature reviews, and theses published between 2014 and 2025. The studies address the parametric design, 3D printing, anthropometric data integration, AI-driven control, functional evaluation, and accessibility of upper limb prostheses and orthoses. A substantial proportion originate from Latin America (particularly Brazil, Colombia, Ecuador, and Chile), with additional contributions from Europe, North America, Africa, and the Middle East. Several sources are duplicates in the dataset (sources 13/39, 45/54, 12/40, and 59/71 share identical abstracts), representing paired entries that are treated as a single study for analytical purposes.

Study	Full text retrieved?	Study Type	Primary Focus
Enzo Romero et al., 2025 [12]	No (abstract only)	Primary study	AI-driven myoelectric transradial prosthesis with 3D printing and BAM-ULA evaluation [12]
Chih Hsing Chu et al., 2022 [20]	No (abstract only)	Primary study	Parametric 3D hand model for customized short thumb orthosis design [20]
Eduardo Keller Rorato et al., 2023 [21]	Yes	Primary study	Open-source aesthetic-functional passive upper limb prosthesis remix [21]
Bárbara Bernadelli Ribeiro et al., 2024 [22]	Yes	Literature review	3D printing in pediatric upper limb prostheses [22]
Raissa Rodrigues Pereira Lima et al., 2019 [23]	Yes	Case report	Functional changes in wrist amputation with customized 3D-printed prosthesis [23]
Millena Serra Souza et al., 2025 [24]	No (abstract only)	Primary study	SLA-printed customized finger orthosis with FEA structural validation [24]
Camilo Andres Solano Rico et al., 2024 [25]	Yes	Primary study	EMG-controlled bionic upper limb prosthesis designed with anthropometric data [25]

Study	Full text retrieved?	Study Type	Primary Focus
Rafael Campos Lima & Francisco Javier de Obaldía Díaz, 2024 [26]	Yes	Primary study	Pediatric EMG-controlled hand prosthesis with tactile simulation [26]
U JorgeZúñiga et al., 2020 [13]	Yes	Case report	Functional and psychosocial impact of low-cost 3D transradial prosthesis [13]
Helder Clay Fares dos Santos Júnior et al., 2024 [14]	Yes	Primary study	Cost-effectiveness of 3D-printed biomechanical orthosis for post-stroke rehabilitation [14]
E. Á. Álvarez Sánchez & Laura Gregorio Falfán, 2019 [27]	Yes	Primary study	3D printing methodology for mechanical hand prosthesis for a child [27]
Alejandro M. Saldarriaga et al., 2024 [28]	No (abstract only)	Primary study	Parametric design approach for affordable customized 3D transradial socket [28]
Rodrigo César da Silveira Romero et al., 2025 [29]	Yes	Primary study	Parametric prostheses for different hand amputation levels via additive manufacturing [29]
Immaculada Llop Harillo, 2020 [30]	No (abstract only)	Thesis	Methods for design and evaluation of anthropomorphic artificial hands [30]
Maria Elizete Kunkel & Mark Barna, 2014 [31]	No (abstract only)	Primary study	Structural modifications and performance of 3D-printed mechanical hand prosthesis [31]
Bruno Bachiega da Silva et al., 2021 [32]	No (abstract only)	Case report	User-centered design and rapid prototyping for transradial prosthesis socket [32]
Silvia SÁEZ OPORTO et al., 2022 [33]	Yes	Primary study	3D technology for digital prostheses in adults with work-related finger amputations [33]
Paola Lizeth Orquiz Avila et al., 2024 [34]	Yes	Primary study	Fit functionality and mechanical properties evaluation of transradial 3D prosthesis [34]

Study	Full text retrieved?	Study Type	Primary Focus
Cristiane Schmitz, 2019 [35]	No (abstract only)	Thesis	Protocol for development and evaluation of 3D customized wrist-hand orthosis for cerebral palsy [35]
Sbtm Cbrm, 2017 [36]	No (abstract only)	Case report	3D assistive technology for manual function deficit in Parkinson's disease [36]
D. Rivera, 2018 [37]	No (abstract only)	Thesis	Design analysis and improvement of a prosthetic hand [37]
André Felipe Cândido da Silva, 2015 [38]	No (abstract only)	Thesis	Robotic hand actuated by shape memory alloy actuators [38]
Zúñiga Guadalupe & Andrés Sebastián, 2018 [39]	No (abstract only)	Thesis	Development and testing of EMG-controlled left arm prosthesis prototype [39]
Vitor Henrique Martio, 2017 [40]	No (abstract only)	Thesis	Myoelectric-controlled hand and wrist prosthesis [40]
Quinga Escobar & P. Antonio, 2018 [41]	No (abstract only)	Thesis	Personalized 3D modeling and creation of index finger prosthesis [41]
João Vitor Leão Silveira Dourado, 2017 [42]	No (abstract only)	Thesis	Fixation and comfort systems for 3D-printed wrist orthoses in cerebral palsy [42]
J. Romero, 2019 [43]	No (abstract only)	Thesis	3D printing-oriented design of prosthetic hand with flexible joints [43]
Cristina Molina Moreno, 2019 [44]	No (abstract only)	Thesis	Open-source 3D-printed arm prosthesis design for congenital malformation [44]
Sbtm Cbrm, 2017a [45]	No (abstract only)	Case report	3D-printed orthosis for ulnar deviation correction [45]
F. Pinheiro & Francisco, 2018 [46]	No (abstract only)	Thesis	Development of functional upper limb prosthesis using flexible filament (TPU) [46]

Study	Full text retrieved?	Study Type	Primary Focus
Tierra Llanga & A. Marcelo, 2017 [47]	No (abstract only)	Thesis	Low-cost myoelectric arm prosthesis for a one-year-old with amniotic band defect [47]
J. Lopez, 2021 [48]	No (abstract only)	Thesis	Accessible 3D-printed hand prostheses using smartphones and non-assembly fabrication [48]
A U Iwuoha & C. Iheme, 2020 [49]	Yes	Primary study	Design and 3D printing of body-powered prosthetic limb for amputee [49]
Kendall F Gretsche et al., 2016 [50]	No (abstract only)	Primary study	Novel 3D-printed robotic shoulder-controlled prosthesis for transradial amputees [50]
A. Elbreki et al., 2022 [9]	No (abstract only)	Primary study	Upper prosthetic limb using 3D printer with AI-based controller (KNN/SVM) [9]
Pascal Weiner et al., 2018 [11]	Yes	Primary study	KIT prosthetic hand: five-finger 3D-printed hand with CNN grasp control [11]
A. Mohammadi et al., 2020 [51]	Yes	Primary study	Paediatric 3D-printed soft robotic hand prosthesis using monolithic TPU design [51]
Julio Fajardo et al., 2017 [52]	No (abstract only)	Primary study	Affordable open-source multifunctional upper limb prosthesis with sEMG control [52]
Rodrigo César da Silveira Romero et al., 2025a [53]	No (abstract only)	Primary study	Parametric prostheses for different hand amputation levels (duplicate entry) [53]
Alejandro M. Saldarriaga et al., 2024a [54]	No (abstract only)	Primary study	Parametric design for affordable 3D transradial socket (duplicate entry) [54]
Daniel Lim et al., 2018 [7]	Yes	Primary study	Customization of 3D-printed prosthetic finger using parametric modeling [7]

Study	Full text retrieved?	Study Type	Primary Focus
J. Zuniga et al., 2015 [1]	Yes	Primary study	Cyborg Beast: low-cost 3D-printed prosthetic hand for children [1]
Julio Fajardo et al., 2020 [2]	Yes	Primary study	Galileo Hand: anthropomorphic and affordable upper limb prosthesis [2]
M. Moreo, 2016 [8]	No (abstract only)	Thesis	Parametric design of 3D-printable hand prosthesis for children in developing countries [8]
Filip Górski et al., 2022 [6]	Yes	Primary study	Automated design and rapid manufacturing of low-cost customized upper limb prostheses [6]
Luz Melissa Chávez Diez de Sollano et al., 2024 [55]	No (abstract only)	Primary study	Upper limb prosthesis prototype with 3D-printed parts and Arduino control [55]
B. Maia, 2016 [56]	No (abstract only)	Thesis	Dimensional parametrization by regression model for children's 3D hand prostheses [56]
I. S. D. Medeiros, 2019 [57]	No (abstract only)	Thesis	User-centered design of upper limb orthosis for child with spastic hemiplegia [57]
R. Moya & Teresa Magal-Royo, 2019 [58]	No (abstract only)	Primary study	Rapid prototyping rehabilitation device for rheumatoid arthritis of the hand [58]
R. Moya & R. Moya, 2019 [59]	No (abstract only)	Primary study	Rapid prototyping rehabilitation device for rheumatoid arthritis (duplicate entry) [59]
R.C.S. Simoes et al., 2018 [60]	No (abstract only)	Primary study	EMG-controlled robotic arm demonstration using open-source InMoov model [60]
V. Leonardo et al., 2018 [61]	No (abstract only)	Primary study	Hydraulically actuated thumb implementation for Poli hand prosthesis [61]

Study	Full text retrieved?	Study Type	Primary Focus
K. Ráž et al., 2024 [5]	Yes	Primary study	Parametric production of forearm prostheses using Multi Jet Fusion technology [5]
Filip Górski et al., 2022a [62]	Yes	Primary study	Automated design and rapid manufacturing of customized prostheses (duplicate entry) [62]
Juan Sebastian Cuellar et al., 2020 [3]	Yes	Primary study	3D-printed hand prosthesis with articulated bio-inspired fingers [3]
J. T. ten Kate et al., 2017 [19]	Yes	Review	3D-printed upper limb prostheses: systematic review of 58 devices [19]
Anna Włodarczyk-Fligier et al., 2025 [63]	No (abstract only)	Primary study	Polymer prosthetic hand with finger copies using 3D scanning and PLA printing [63]
T. D. Stocco & Renan Alves Rodrigues, 2020 [17]	Yes	Literature review	Use of 3D-printed prosthetic limbs for amputees: literature review [17]
Letícia Alcará da Silva et al., 2018 [64]	Yes	Case report	Interdisciplinary development of customized 3D-printed upper limb prosthesis [64]
Daniel De Castro Maciel et al., 2024 [65]	Yes	Primary study	Prosthesis development for absent medial phalanx [65]
Júlia Meireles Boechat et al., 2023 [66]	No (abstract only)	Primary study	Transradial myoelectric bionic prosthesis for adults with traumatic amputation [66]
Beatriz Paredes do Nascimento et al., 2024 [67]	Yes	Primary study	Additive manufacturing for upper limb prosthesis production: case study [67]
Caio Lima et al., 2020 [68]	No (abstract only)	Primary study	Force threshold detection in forearm prosthesis with resistive sensors [68]
Isabela Victória Fontes Arizi et al., 2020 [69]	Yes	Case report	Dexterity, motor coordination, and quality of life evaluation with assistive technology [69]

Study	Full text retrieved?	Study Type	Primary Focus
Mariana Francisco Ramos et al., 2021 [70]	No (abstract only)	Literature review	Upper limb orthoses for people with cervical spinal cord injury [70]
Yongzhao Xu et al., 2022 [10]	No (abstract only)	Primary study	Intelligent EMG pattern recognition with artificial neural network for prosthesis control [10]
Moosa S. M. Al Kharusi et al., 2024 [71]	No (abstract only)	Primary study	AI-driven design of customized prosthetic limbs with additive manufacturing [71]
Vladimir Cruz Vilchis et al., 2024 [72]	Yes	Primary study	Biomechanical forearm prosthesis prototype with PETG and EMG control [72]
R. Caselas, 2018 [73]	No (abstract only)	Thesis	Materialization of imaginary through aesthetics of upper limb prostheses [73]
C. Oliveira, 2021 [74]	No (abstract only)	Primary study	Automation of a 3D-printed hand prosthesis with Arduino and EMG [74]
Letícia Alcará da Silva et al., 2018a [75]	Yes	Case report	Interdisciplinary customized 3D upper limb prosthesis (duplicate entry) [75]
I. Vujaklija & D. Farina, 2018 [76]	No (abstract only)	Review	3D-printed upper limb prosthetics: current solutions and expert commentary [76]
Yunus Emre Kahraman et al., 2020 [77]	No (abstract only)	Primary study	Parametric design and modeling of 3D-printed prosthetic finger [77]
Abdalla M. Omar & Mohamed H. Hassan, 2020 [78]	Yes	Primary study	Design of 3D-printed smart material compatible hand prosthesis [78]
Albert Manero et al., 2019 [16]	Yes	Review	Implementation of 3D printing technology in the field of prosthetics: past, present, future [16]
A. Mohammadi et al., 2020a [15]	Yes	Primary study	Practical 3D-printed soft robotic prosthetic hand with multi-articulating capabilities (X-Limb) [15]

Study	Full text retrieved?	Study Type	Primary Focus
Said Dababneh et al., 2025 [4]	No (abstract only)	Review	Shaping the future of upper extremity prostheses through 3D printing [4]
Christian Andres Diaz Leon, 2019 [79]	Yes	Thesis	Application of 3D additive manufacturing and robotics for hand prosthetic design [79]
V. Bharath et al., 2023 [80]	No (abstract only)	Primary study	Anthropomorphic upper extremity design based on EMG using 3D technology [80]
K. Wendo et al., 2022 [18]	Yes	Review	Open-source 3D printing in the prosthetic field: upper limb prostheses review [18]

The corpus spans a wide methodological range: reviews synthesizing dozens of existing devices, prospective case studies with formal outcome measurement, engineering theses documenting prototype development, and retrospective case reports. Eleven sources are full reviews or systematic reviews of the field, while the remainder describe original device development or clinical evaluation. This heterogeneity must be kept in mind when interpreting the breadth versus depth of evidence for any specific sub-question.

3D Printing Technologies and Materials

Technology Distribution

Fused Deposition Modeling (FDM), also referred to as Fused Filament Fabrication, is by far the dominant technology across this body of literature. Among sources that specify a technology, the overwhelming majority rely on FDM [1, 3, 6, 7, 15, 17, 18, 29, 34, 49, 64, 65, 69, 78]. The review by ten Kate et al. (2017), which catalogued 58 devices, reported that 46 were manufactured with FDM, six with SLS, one with SLA, and one with polyjet printing [19]. Stereolithography (SLA) appears only in the orthosis study of Souza et al. (2025), which used an Anycubic Photon printer with photopolymerizable PrintaX Blue-405 resin [24]. Multi Jet Fusion (MJF) represents the most technically sophisticated alternative documented in the set, employed by Ráž et al. (2024) using HP MJF technology with PA12GB (polyamide reinforced with 40% glass beads) at a layer thickness of approximately 0.08 mm [5]. The KIT prosthetic hand by Weiner et al. (2018) used Selective Laser Sintering (SLS) with PA2200 nylon powder [11], a choice that eliminates the need for support structures and yields superior surface quality compared to FDM.

Materials

PLA (polylactic acid) is the most frequently mentioned printing material, cited across at least 20 studies, and is favored for its biocompatibility, ease of printing, and low cost [1, 7, 9, 14, 17, 23, 25, 28, 29, 34, 39, 55, 63–67, 78, 80]. ABS appears in several earlier studies [1, 13, 17, 31, 38, 41, 47, 69] but is associated with printing difficulties such as

warping, harmful gas emission during printing, and poorer inter-layer bonding compared to PLA [34, 69]. Flexible materials, particularly TPU (thermoplastic polyurethane), are used where compliance and soft contact are desired: Mohammadi et al. (2020) used TPU90 for a monolithic pediatric hand [51], Mohammadi et al. (2020a) used the same material for the X-Limb prosthesis [15], Pinheiro & Francisco (2018) used flexible TPU filament for a prosthetic hand designed around compliance [46], and Cuellar et al. (2020) employed a multi-material approach combining PLA, TPU, and Nylon to fabricate bio-inspired articulated fingers [3]. PETG was selected by Cruz Vilchis et al. (2024) for a forearm prosthesis prototype [72], and Tritan—a food-safe copolyester—was used by Romero et al. (2025) for its hypoallergenic properties [29]. Thermoplastic elastomers (TPE) and thermoplastic polyurethane are used in some cases where skin contact comfort is prioritized [79].

Printing Parameters and Hardware

Among the few studies that report detailed parameters, the Lim et al. (2018) study specified a layer thickness of 0.15 mm, print speed of 60 mm/s, and infill density of 20% on an Ultimaker 2.0+ [7]. Omar & Hassan (2020) used a layer height of 0.15 mm, infill density of 15% for the hand, and print speed of 60 mm/s [78]. Mohammadi et al. (2020a) reported extrusion speed of 60 mm/s, travel speed of 80 mm/s, nozzle temperature of 220°C, and infill density of 30% for fingers and 20% for the palm, using a hexagonal infill pattern on a FlashForge Guider II [15]. Iwuoha & Iheme (2020) specified bed temperature of 20–60°C, nozzle temperature of 210–230°C, and print speed of 30 mm/s for TPU on a MakerBot printer [49]. Reporting of printing parameters is the exception rather than the rule, which limits reproducibility across the literature.

Manufacturing times when reported range from approximately 6 hours for a single finger orthosis [24] and PLA prosthetic components [29], to 20–28 hours for complete prosthetic assemblies [6, 14]. The TPE hand by Diaz Leon (2019) required 23 hours of printing time [79]. Post-processing is minimal for most FDM devices, typically limited to filing, sanding, and assembly [49, 64], while SLS parts may require polishing or coating [19].

Anthropometric Data Collection and Parametric Design

Measurement Methods

The studies employ a spectrum of anthropometric data collection strategies ranging from manual tape-measure assessment to structured photogrammetry and 3D laser scanning. Manual measurement is the most prevalent approach, typically capturing limb length, circumferences at multiple points, and stump dimensions, which are then entered into CAD parametric models [13, 23, 27, 65, 67, 72]. Zuniga et al. (2015) validated distance measurement via photographs taken from standardized positions, finding no significant bias for length and width measurements and acceptable limits of agreement [1]; this distance-fitting methodology was specifically developed to address the inaccessibility of in-person prosthetists in developing countries. Lopez (2021) extended this concept by attempting smartphone photogrammetry combined with a Statistical Shape Model (SSM) to derive socket dimensions automatically, but found the resulting measurement errors too large for clinically acceptable socket fit, ultimately resorting to manual measurement with a tape measure [48].

3D scanning is reported in a substantial subset of studies. Górski et al. (2022) employed a David SLS-3 optical scanner to capture residual limb geometry, feeding the output through MeshLab and into Autodesk Inventor via iLogic rules linked to an Excel spreadsheet [6, 62]. Ráž et al. (2024) used an automatic laser projector with a scan resolution of 0.2 mm and allowed a maximum surface deviation of 1 mm between the scanned model and the generated parametric socket [5]. Włodarczyk-Fligier et al. (2025) used an EinScan PRO HD SHINING 3D scanner combined with Autodesk Inventor and Meshmixer [63], and the REACH group (Simoes et al., 2018) similarly employed 3D scanning to capture

patient geometry before printing [36]. Schmitz (2019) used 3D scanning as the acquisition step within a protocol subsequently modeled in Meshmixer [35].

Chu et al. (2022) collected 19 anthropometric parameters from 120 Taiwanese adults using a scanning device and constructed a statistical hand parametric model; individual client values were then imported to synthesize a personalized mesh for orthosis fabrication [20]. Maia (2016) took a regression modeling approach, seeking to establish a single predictive parameter—stature—that could estimate hand length and consequently drive prosthesis dimensioning, a solution specifically targeting the scarcity of pediatric anthropometric data [56, 56]. Moreo (2016) conducted a statistical analysis revealing that four key parameters—grip circumference, palm breadth, thumb breadth, and palm length—could all be derived from a single driver variable, palm breadth, enabling a one-parameter parametric design workflow [8]. Lim et al. (2018) demonstrated through measurements from 50 adults that finger length and width are negligibly correlated, meaning uniform scaling of a prosthetic finger cannot simultaneously match both dimensions for a given user; the parametric model they proposed adjusts length and width independently, yielding a range of motion path that closely matches the natural digit [7, 7].

Integration with CAD Systems

The CAD environments reported include SolidWorks [24, 25, 34, 65, 78, 79], Autodesk Fusion 360 [23, 25, 49], Autodesk Inventor [6, 62, 63], Rhinoceros with Meshmixer [64, 75], Blender [1, 21, 69], and OpenSCAD [67]. Parametric workflows that link spreadsheet-based patient data directly to geometry drivers in the CAD environment are the most sophisticated approach documented, as implemented by Górski et al. (2022) with Inventor/iLogic/Excel [6] and Ráž et al. (2024) with their VXelements-to-STL pipeline [5]. The prosthesis by Romero et al. (2025) used iLogic constraints in a CAD environment and achieved printing in approximately 6 hours at a cost of USD 50.15, compared to traditional prosthetics ranging from USD 254.74 to USD 984.26 [29].

AI and Control System Integration

EMG-Based Control

The most widely implemented form of AI or automated intelligent control in this literature is surface electromyography (sEMG) signal processing for prosthesis actuation. Multiple studies developed EMG-based systems to capture voluntary muscular contractions and translate them into prosthetic movements. Romero et al. (2025) employed machine learning for real-time EMG classification, achieving gross motor task success rates of 100% and fine motor task rates of 22.2% on the BAM-ULA evaluation [12, 12]. Elbreki et al. (2022) tested K-Nearest Neighbor and Support Vector Machine classifiers on an extracted feature database of 416×4 observations, achieving 94% accuracy with KNN and 95% with SVM across four movement classes [9, 9]. Xu et al. (2022) developed an artificial neural network system trained on signals from 20 volunteers and validated with 15 new participants, implementing the full pipeline via a web application with cloud-hosted database [10]. The REACH project (Simoes et al., 2018) demonstrated voluntary control of a robotic arm using an EMG shield and Arduino UNO, with signals from the common flexor muscles driving five servo motors via polyamide cables [60]. Cruz Vilchis et al. (2024) used a Raspberry Pi Pico to interpret EMG signals and actuate a servomotor for forearm prosthesis grip [72]. Tierra Llanga & Marcelo (2017) reported a system reaction time of 2.3 seconds to a child's signals and 0.26 seconds to adult muscle signals [47], demonstrating that pediatric muscle signal amplitude poses a distinct challenge.

The KIT prosthetic hand (Weiner et al., 2018) represented the highest level of AI integration in the set, embedding a Convolutional Neural Network (CNN) trained on 300 images per household object for context-dependent grasp selection, combined with an RGB camera in the palm base and a color display on the dorsum [11]. The hand achieved

a finger-tip force of 7.48–11.82 N, hook grasp force of 120 N, and hand closing time of approximately 1.3 seconds [11]. The Galileo Hand (Fajardo et al., 2020) used a graphical user interface (μ LCD screen) in combination with sEMG electrodes, achieving MCP joint flexion and extension times of 800 ms and 600 ms respectively, thumb abduction and adduction times of 150 ms, and a maximum per-finger load of 2.5 kg [2]. The NASA Task Load Index experiment indicated low physical and temporal demands and minimal frustration from users [2].

André Felipe Cândido da Silva (2015) implemented a fuzzy logic controller to monitor phalange angles in a shape-memory alloy-actuated robotic hand, partially achieving target angles [38, 38]. Lima et al. (2020) used resistive fingertip sensors and the commercial Myo armband to enable force-threshold grip control, successfully grasping fragile objects such as disposable cups without crushing them [68].

Statistical Shape Models and Automated Parametrization

Lopez (2021) implemented a Statistical Shape Model to automate anthropometry from smartphone photographs, though error rates proved too high for clinical socket design without manual correction [48]. Górski et al. (2022) used intelligent CAD models with adaptive geometry that automatically reconfigured in response to 3D scan inputs, representing a form of rule-based automated parametrization short of machine learning [6, 62]. Ráž et al. (2024) similarly used a special algorithmic model to generate editable parametric sockets from scan data [5]. Maia (2016) applied regression modeling to create a single-parameter lookup table from stature to hand dimensions [56]. Chu et al. (2022) used a population of 120 adult scans to build a statistical parametric hand model from which personalized geometries could be synthesized by inserting 19 measured values [20].

Functional Performance

Grip Strength and Mechanical Performance

Quantitative mechanical data are reported by a subset of studies. The X-Limb (Mohammadi et al., 2020a) demonstrated a power-grip force of 21.5 N and finger flexion speed of 1.3 seconds, with a durability test of at least 45,000 cycles while maintaining original functionality [15]. The KIT hand achieved finger-tip forces of 7.48–11.82 N and a hook grasp of 120 N [11]. Cuellar et al. (2020) measured an energy dissipation of 0.380 Nm per cycle and a pinch force of approximately 16 N at an input force of 100 N, superior to other fully 3D-printed body-powered designs but below conventional body-powered prostheses [3]. Górski et al. (2022) subjected their mechanical prosthesis to structural testing, recording break forces of 500 N at finger joints, with fingers sustaining up to 1000 N and the palm up to 1200 N of bending force [6, 62]. Moreo (2016) reported that the parametric prosthetic hand could provide pinch force up to 35 N for a 13-year-old child, above the minimum 10 N requirement [8]. Solano Rico et al. (2024) recorded a maximum weight capacity of 1.5 kg with a 14.7 N force and 0.2283 mm displacement under load [25]. Cruz Vilchis et al. (2024) noted acceptable performance at 10–60 kg of applied force but difficulty above 60 kg [72]. Omar & Hassan (2020) reported mechanical strength up to 500 N in uniaxial tension tests of a 22 degrees-of-freedom hand with PLA Plus, which begins to yield at 538 N [78].

Standardized Clinical Assessment

Several studies employed standardized outcome measures. Romero et al. (2025) used the BAM-ULA (Brief Activity Measure for Upper Limb Amputees), yielding an average score of 7.4/10 (SD 0.7) across nine participants, with 100% success on gross motor tasks and 22.2% on fine motor tasks [12]. Mohammadi et al. (2020a) applied the AM-ULA (Activities Measure for Upper-Limb Amputees) benchmark and reported that all practical grasping tasks were

successfully performed [15]. Zuniga et al. (2020) employed the Bilan 400 points modified and the Upper Extremity Functional Index (UEFI), reporting approximately 30% improvement in hand functionality over an 11-month follow-up [13]. Sáez Oporto et al. (2022) used the Bilán test, DASH, and SF-36 over a 6-month period with eight adults following digital amputation from occupational accidents, finding significant improvements in hand functionality and reductions in disability perception, though one participant withdrew [33, 33]. Fares dos Santos Júnior et al. (2024) measured Medical Research Council (MRC) muscle strength in eight post-stroke patients, recording improvement from 2.62 ± 0.51 to 3.75 ± 0.46 following use of the 3D-printed biomechanical elbow orthosis [14]. Schmitz (2019) used the Jebsen-Taylor Test Hand Function, recording improvements of 13 seconds in writing speed, 0.9 seconds in picking up small objects, and 69.3 seconds in simulated feeding [35]. Manero et al. (2019) documented a one-year clinical trial in a pediatric population using the Assessment of Capacity for Myoelectric Control (ACMC), the Children's Hand-Use Experience Questionnaire (CHEQ), and PedsQL [16].

Task-Based and Informal Evaluation

The majority of studies report functional outcomes through less structured means. Kunkel & Barna (2014) demonstrated that their ABS prosthesis enabled a bilateral amputee to drink from a cup and write their name [31]. Lima et al. (2019) recorded specific Action Research Arm Test score changes: grip (squeezing) +2 points, holding +1.5 points, gross movement +3 points, but no gain on pinch (0 points) because the prosthesis only achieves power grasping [23]. Álvarez Sánchez & Gregorio Falfán (2019) progressed from a 147-gram bottle lift to a 517-gram lift across two prototype versions [27]. Maciel et al. (2024) demonstrated that a phalangeal prosthesis achieved flexion angles closely matching normal human finger angles and enabled typing and object grasping [65]. Iwuoha & Iheme (2020) reported a prosthesis weight of 188 g (below the 200 g target), with comfortable socket fit and no painful pressure on residual limb bands [49]. The Cyborg Beast (Zuniga et al., 2015) was evaluated in 13 children over one to three months; four children reported significant quality of life increases, seven small increases, and one no change, with nine using it one to two hours daily [1].

User Experience and Acceptance

Satisfaction and Psychosocial Outcomes

User experience data are fragmented across the corpus, though consistently positive where reported. Zuniga et al. (2020) documented PIADS (Psychosocial Impact of Assistive Devices Scale) subscores for competence, self-esteem, and adaptability rising from zero effect to a maximum of 3 points for adaptability by 11 months [13]. Average daily use in this case was 5 hours [13]. The patient continued use without abandonment through the 11-month follow-up [13]. The Quebec User Evaluation of Satisfaction with Assistive Technology scores were mostly high for the Chu et al. (2022) thumb orthosis, and the orthosis was described as providing greater flexibility and stronger support than the manually-formed traditional equivalent [20, 20]. Silva et al. (2018) reported the pediatric user was delighted with the result, describing comfort, ease of placement via Velcro, and satisfaction with the aesthetic personalization featuring the child's preferred characters and colors [64, 75].

The study by Arizi et al. (2020) provides an important cautionary finding: the e-NABLE Cyborg Beast design was not satisfactory for a patient with congenital transradial amputation, whose musculature differs substantially from traumatic amputees for whom the device was originally designed [69]. This contrasts with the positive reports for the same design in other populations, illustrating that etiology of limb absence (congenital versus traumatic) is a critical moderator of user satisfaction.

Sbtm Cbrm (2017) documented improved performance in activities of daily living using a 3D-printed universal adapter for spoon, toothbrush, and crochet needle use in a Parkinson's patient, with a mean occupational performance score of 6.0 and satisfaction score of 4.5 on the Canadian Occupational Performance Measure [36, 36]. Sbtm Cbrm (2017a) reported ulnar deviation correction from 6°–10° to 0° in all three patients, pain reduction (EAV scores: patient S1 from 7/10 to 4/10, S2 from 6/10 to 2/10, S3 from 8/10 to 0/10), and subjective satisfaction with lightness and washability of the 3D-printed orthoses [45, 45].

Aesthetic Personalization

Several studies highlight color, shape, and character-based personalization as meaningful to users, especially children. Romero (2019) and Caselas (2018) both discuss how additive manufacturing enables aesthetic expressions trending toward a "cyberpunk" or super-human aesthetic rather than traditional skin-toned cosmesis, and that users often prefer transparent "robot" aesthetics over naturalistic ones [73]. Eduardo Keller Rorato et al. (2023) explicitly documented that users favor aesthetic and social use of passive prostheses over purely functional mechanisms [21]. The orthosis literature (Schmitz, 2019; Medeiros, 2019) similarly notes that aesthetic appeal contributes to improved adherence and reduced abandonment [35, 57].

Abandonment

Manero et al. (2019) cites pediatric rejection rates for upper limb prostheses ranging from 38% to 45% in the traditional prosthetic literature, and identifies weight, perceived durability, and aesthetics as primary drivers [16]. Dourado (2017) developed fixation and comfort systems specifically to reduce orthosis abandonment by eliminating pressure points between the hand and the orthosis [42]. The e-NABLE community's ecosystem—offering free designs, color customization, and community support—is noted by multiple reviews as addressing rejection through personalization [16, 19].

Cost and Accessibility

Material costs for 3D-printed upper limb prostheses span an exceptionally wide range, broadly clustered at the low end of prosthetic pricing. Zuniga et al. (2015) reported a material cost of approximately USD 50 for the Cyborg Beast, compared to traditional myoelectric prostheses costing USD 4,000–20,000 [1]. The Galileo Hand cost less than USD 350 to build [2]. Cuellar et al. (2020) achieved a material cost of USD 12 for the bio-inspired hand [3]. The parametric prosthesis by Romero et al. (2025) cost USD 50.15 per unit against traditional alternatives ranging from USD 254.74 to USD 984.26 [29]. Górski et al. (2022) reported costs not exceeding USD 250 for a complete customized prosthesis and USD 75 for a mechanical-only prosthesis [6]. Mohammadi et al. (2020a) reported a bill-of-materials cost of approximately USD 200 for the X-Limb [15]. The finger orthosis by Souza et al. (2025) was produced for less than R\$6.00 (approximately USD 1.20) [24]. At the upper bound of the 3D-printed category, the KIT hand carried a material cost of approximately €1,000, with motors alone accounting for €474 [11]. Gretsche et al. (2016) estimated their robotic shoulder-controlled design at USD 300 [50]. In the Dababneh et al. (2025) review of 35 prostheses, 64% of models cost under USD 50 [4].

Production time reductions are consistently reported against traditional laminate composite or thermoplastic custom fabrication. Górski et al. (2022) reported complete prosthesis production in approximately 20 hours for one patient and 13 hours for another [6]. Ráž et al. (2024) emphasized that the MJF workflow dramatically accelerates delivery compared to traditional long-fiber composite lamination, which produces only one socket at a time and requires multi-day curing [5]. The Cyborg Beast requires approximately 2.5 hours for assembly after printing [1]. Stocco

& Rodrigues (2020) noted forearm prosthesis costs in the range of USD 20–1,250 for 3D-printed devices, far below commercial equivalents [17].

Geographic and economic accessibility are recurrent themes, particularly for work from Latin American, African, and Middle Eastern groups. The Tierra Llanga & Marcelo (2017) myoelectric prosthesis for a one-year-old cost USD 800 against a comparable commercial device priced at USD 29,379—a 97.3% cost reduction [47]. The REACH project (Simoes et al., 2018) explicitly aimed at distributing prostheses through Brazil's public health system (SUS) [60]. Nascimento et al. (2024) estimated a complete prosthetic device at R\$115.00, approximately 2% of the market value of a commercial aesthetic prosthesis [67]. Lopez (2021) specifically designed a prosthesis that could be manufactured with a single FDM printer in field conditions in Colombia, demonstrating that the design and manufacturing process is fast and easy to implement in developing countries [48].

Assembly difficulty is a practical accessibility concern. The Dababneh et al. (2025) review rated the majority of reviewed designs as easy to moderate in assembly, making them accessible for non-specialist users [4]. The e-NABLE community has been particularly influential in disseminating assembly-ready designs globally [16].

Technical Limitations

Material Durability

Material durability is the most consistently cited technical limitation across the corpus. PLA—despite being the most popular material—has documented sensitivity to elevated temperatures [34, 65], moisture exposure, and mechanical fatigue. Stocco & Rodrigues (2020) describe 3D-printed prosthetics as having less durable materials, lower rigidity, and poor thermal stability relative to traditional alternatives [17]. Ten Kate et al. (2017) note that predicting life cycles for printed parts is difficult and that fractures and wear require users to obtain reprints [19]. Mohammadi et al. (2020a) observed membrane cracks in the X-Limb TPU structure after the durability test, and noted that cosmetic gloves add stiffness that reduces grip force [15]. Moreo (2016) identified durability as the primary unresolved challenge for the parametric children's prosthesis [8]. Manero et al. (2019) reported that the mechanical properties of ABS vary by print orientation from 10% to 73% of injection-molded values, a critical source of performance variability [16].

Grip Strength and Fine Motor Function

Grip strength is consistently lower in 3D-printed body-powered and passive devices than in conventional alternatives. Gretsche et al. (2016) explicitly identified low grip strength and decreased durability as limitations relative to passive prosthetics [50]. Cuellar et al. (2020) acknowledged that their bio-inspired design's pinch force—while superior to other fully printed body-powered designs—remains below conventional body-powered prostheses [3]. Romero et al. (2025) found fine motor task completion rates of only 22.2%, compared to 100% for gross motor tasks, underscoring the persistent gap between coarse and fine manipulation [12]. Lima et al. (2019) similarly found no functional gain on the pinch subscale of the Action Research Arm Test, as the prosthesis only achieved gross prehension [23].

Uniform Scaling Versus True Parametrization

Lim et al. (2018) demonstrated quantitatively that uniform scaling—the approach used by the majority of open-source prosthetic designs—produces range-of-motion paths that diverge from the natural digit trajectory when length-to-width ratios differ from the baseline model [7]. Among their 50-participant sample, 52% exhibited length differences exceeding 5% when fitted with a uniformly scaled design [7]. This finding is methodologically important: it suggests

that many studies claiming “customized” 3D-printed prostheses are implementing only simplified scaling rather than genuine anthropometric parametrization.

Socket Fit and Measurement Accuracy

Achieving consistent socket fit is a recurring challenge. Lopez (2021) found that smartphone photogrammetry introduced errors too large for acceptable socket design [48]. Orquiz Avila et al. (2024) noted discrepancies between EBSCOhost-sourced anthropometric data and SolidWorks-generated dimensions [34]. The Górski et al. (2022) approach required manual correction of coordinate systems in scan data [6]. Ráž et al. (2024) imposed a tolerance of 1 mm between the scanned and generated surfaces as an acceptability criterion [5]. Multiple studies report that socket fitting required iterative physical adjustment following initial printing [64, 75], indicating that purely computational approaches to socket design have not yet eliminated clinical refinement steps.

Design Complexity and Regulatory Context

Wendo et al. (2022) and Dababneh et al. (2025) both highlight the lack of standardized manufacturing protocols and limited clinical validation as systemic challenges [4, 18]. Ten Kate et al. (2017) note the absence of a regulatory and standards framework specifically for 3D-printed prostheses, and the difficulty of predicting mechanical property equivalence with traditionally manufactured devices [19]. Manero et al. (2019) specifically discusses FDA part-tracking and documentation requirements as a challenge for maker-community devices that operate outside conventional prosthetic approval pathways [16]. Stocco & Rodrigues (2020) identify the need for more qualified professionals to operate additive manufacturing equipment in clinical prosthetic workflows [17].

Synthesis

Apparent Contradictions in the Evidence

The literature presents an apparent contradiction: a large number of studies report successful functional outcomes and high user satisfaction with 3D-printed personalized prostheses and orthoses, yet reviews and technical analyses consistently identify durability, grip strength, and clinical validation as unresolved problems. This tension can be explained by examining several contextual distinctions.

Population and intended use distinctions. The devices achieving highest reported satisfaction and functional improvement are predominantly passive or body-powered prostheses intended for children in low-resource settings who otherwise have no prosthesis at all [1, 8, 16, 67, 69]. For this population, even a device with modest grip strength and limited durability represents a transformative functional and psychosocial gain [13, 16]. In contrast, studies targeting adults with traumatic amputation and comparing with commercial myoelectric devices consistently identify the gap in grip force and fine motor capability [3, 12, 50]. Both conclusions are correct within their respective contexts: 3D-printed devices are functionally adequate relative to no device, but functionally inferior relative to commercial myoelectric prostheses.

Parametrization depth moderates outcomes. Studies implementing genuine anthropometric parametrization—where multiple independent geometric drivers are adjusted from patient measurements—achieve superior fit and user satisfaction compared to uniform scaling approaches [6, 7, 7, 8]. The KIT hand [11], the X-Limb [15], the Galileo Hand [2], and the Romero et al. (2025) transradial system [29] represent the highest technical sophistication in this regard, combining parametric anthropometric fitting with electronic or AI-based control. Simpler open-source designs using uniform scaling achieve lower grip forces and less optimal range of motion, particularly when hand

dimensions deviate from the design baseline [7]. This explains why studies reporting functional equivalence with commercial devices tend to be those implementing multidimensional parametrization and electronic actuation, while those reporting low grip strength tend to involve body-powered or uniformly scaled designs.

AI control matures grip capability. EMG-based systems with machine learning classifiers (SVM, KNN, ANN) consistently achieve classification accuracies in the 90–95% range [9, 10], but this accuracy is measured in controlled laboratory conditions and does not automatically translate to robust clinical performance during daily activities. The gap between fine motor task performance (22.2% success in Romero et al.) [12] and motion classification accuracy (94–95%) [9] reflects the distinction between signal decoding accuracy and actual task completion, which involves additional mechanical constraints of the prosthesis itself. CNN-based grasp selection (KIT hand) [11] attempts to bridge this gap by automating grasp type selection rather than relying solely on user effort, but introduces hardware complexity and cost that run counter to the accessibility mandate driving most of this literature.

Measurement method determines socket quality. Studies achieving the best socket fit are those using 3D scanning with laser projectors and algorithmic parametric socket generation with sub-millimeter tolerances [5, 6]. Manual measurement with a tape measure, while sufficient for hand prosthesis gross dimensioning, introduces variability that propagates into socket comfort [48]. Photogrammetric approaches remain premature for socket design without significant algorithmic refinement [48]. This means that the scalability advantage of remote fitting—enabling prosthesis delivery to rural or resource-limited populations—is currently in tension with the fit quality achievable without in-person 3D scanning.

Cost reduction is robust across the literature. Unlike the functional performance evidence, the cost advantage of 3D-printed prostheses is consistent regardless of device type, population, or geographic setting. Material costs range from less than USD 2 to USD 1,000 across devices, with the vast majority falling between USD 50 and USD 350 [1–4, 6, 15, 50]. Even the high end of this range is one to two orders of magnitude below commercial myoelectric prostheses. This cost advantage is not offset by inferior durability in populations without access to commercial alternatives, since some prosthesis is categorically superior to none. The durability argument has greatest force when 3D-printed devices are positioned as replacements for—rather than supplements to—commercial prosthetics in patients who have access to both.

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